

CIRCUS: A Causal Intervention-Based Framework for Enhancing Counterfactual Fairness in Trained Classifiers

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Abstract—Ensuring model fairness for preventing potential biases based on any sensitive attribute is crucial for the societal acceptance of artificial intelligence in critical applications. Among various fairness concepts, counterfactual fairness has gained prominence as it is grounded in causal inference. This concept requires that an individual’s prediction in the original world remains consistent with that in the counterfactual world where the sensitive feature value is modified. In this article, we aim to mitigate counterfactual biases of the model through causal intervention. Specifically, we first achieve effective causal intervention and counterfactual generation by proposing the causal inference tabular generative adversarial network (CITGAN) architecture. Unlike prior approaches based on variational autoencoders (VAEs) that inherently lack structural causal model (SCM) fidelity due to simultaneous generation, CITGAN strictly enforces causal consistency via an end-to-end topological generation process. By integrating exogenous variable inference with sequential generation, CITGAN ensures that functional dependencies are structurally preserved. Building on CITGAN, we propose the CIRCUS framework, a causal intervention-based framework designed to intuitively enhance the counterfactual fairness in trained classifiers. CIRCUS generates counterfactually discriminatory samples (CDSs) via causal intervention, guided by gradients and feature contributions, and subsequently applies bias correction preprocessing to their labels for classifier retraining. Experimental results demonstrate that CIRCUS effectively enhances counterfactual fairness while maintaining robust classification performance. Specifically, for the deep neural network (DNN) model, the MMD_L and MMD_K values are reduced by averages of 39.7% and 40.4%, respectively, compared with the second-best result. For the residual network (ResNet) model, these reductions amount to 56.7% and 54.5%, respectively.

Index Terms—Causal intervention, causal relation, counterfactual fairness, deep neural network (DNN).

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I. INTRODUCTION

AS ARTIFICIAL intelligence plays an increasingly critical role in fields such as hiring, college admissions, credit services, healthcare, and criminal justice, concerns regarding the potential risks associated with these decision models have intensified. In particular, the issues of unfairness toward protected social groups have attracted considerable attention. For example, research has shown that an algorithm used in US hospitals for medical service allocation systematically discriminates against Black individuals [1]. These inequities, which disproportionately harm the interests of specific demographic groups, have raised significant concerns among society and government agencies. Fairness, as a fundamental ethical guideline for artificial intelligence, requires practitioners to ensure machine-learning (ML) systems do not discriminate against individuals or groups based on legally sensitive features [2].

To evaluate decision bias, scholars have proposed many concepts of fairness. These fairness concepts include individual fairness and group fairness [3]. Recent years have witnessed significant attention from researchers toward causality-based fairness. Causal fairness frames unfairness as the presence of unjust causal effects from sensitive features on decision-making. Building on Pearl’s structural causal model (SCM) [4], several causal fairness concepts have been proposed. This article focuses on counterfactual fairness [5], a concept of causal fairness applied at the individual level. Counterfactual fairness requires that the decision outcome for an individual in the real world remains consistent with the outcome in the counterfactual world. In this framework, individuals in the real world and their counterparts in the counterfactual world form counterfactual pairs. Counterfactual fairness can identify disparities caused by causal factors versus genuinely unfair decisions, providing a more meaningful criterion than group fairness. For instance, in the Berkeley graduate admissions case [6], the lower admission rate for women was due to their applying to departments with lower acceptance rates, rather than gender-based discrimination given equal qualifications. Efforts to raise female admissions to promote group fairness could potentially have adverse effects on male applicants. A common technique to improve counterfactual fairness in models is the introduction of a counterfactual fairness penalty regularization term into the loss function during model training [7], [8], [9], [10]. Some studies [11], [12] use counterfactual augmentation methods, which generate counterfactually fair representations (CFRs) from the original dataset and combine them with the original dataset to create a more

balanced training set for improving counterfactual fairness of models.

However, most existing methods for improving counterfactual fairness face two major limitations. First, their counterfactual reasoning frameworks typically generate counterfactual data using decoders that ignore causal dependencies among endogenous variables. Existing generative methods, particularly those based on variational autoencoders (VAEs), infer exogenous variables and generate endogenous features simultaneously. However, this approach lacks SCM fidelity, as it fails to explicitly enforce the causal dependencies in the data mechanism. As a result, the generated counterfactual examples may be causally inconsistent and less reliable. Second, these methods often lack intuitive guidance and cannot clearly reveal the dynamic changes in prediction results caused by variations in specific features, and therefore often appear as closed-box frameworks with limited interpretability.

To address the first challenge, this article presents the causal inference tabular generative adversarial network (CITGAN). In contrast to prior counterfactual reasoning frameworks, CITGAN explicitly infers exogenous variables from original data and models structural equations according to the topological order of the given causal graph, in which generated parent nodes serve as direct inputs to child generators. This design ensures that synthesized data strictly adheres to the underlying causal logic, thereby guaranteeing both interpretability and SCM fidelity. To better adapt CITGAN to the structure of tabular datasets and reduce the impact of column order, we incorporate the Fourier network blocks (FNBs) module [13] for causal data generation. To address the second challenge, we design a scalable causal intervention-based counterfactual fairness (CIRCUS). Unlike existing counterfactual fairness methods that often behave like closed boxes, CIRCUS is explicitly designed to generate counterfactually fair samples. It does so by first leveraging CITGAN to perform causal interventions on selected features at both the global and local stages, producing counterfactually discriminatory samples (CDSs) that remain consistent with the underlying causal relationships. Subsequently, CIRCUS adjusts the labels of these samples using bias correction preprocessing to yield counterfactually fair samples. The main contributions of this article are summarized as follows.

- 1) We propose a novel architecture named CITGAN to enable effective causal intervention and counterfactual generation. Unlike existing counterfactual reasoning frameworks, CITGAN integrates the inference of exogenous variables with causally ordered data generation, thereby simulating the complete counterfactual reasoning process with high SCM fidelity.
- 2) We design a novel framework named CIRCUS to intuitively generate CDSs through causal intervention at global and local stages, and then apply bias correction preprocessing to convert them into counterfactually fair samples, thereby improving classification fairness while retaining interpretability.

II. RELATED WORK

A. Counterfactual Fairness in ML

With the rise of causal fairness, counterfactual fairness has emerged as a key research topic. It evaluates fairness at the individual level using counterfactual pairs derived from counterfactual reasoning, commonly implemented through encoder-decoder structures [8], [11], [12], [14], [15]. Within this architecture, the encoder's hidden representations serve as exogenous variables, and the decoder predicts counterfactual data based on these representations and the sensitive features. These methods typically assume independence between exogenous and observed variables. The encoder approximates the posterior of the exogenous variables and regularizes it toward a predefined prior, enabling their inference from observational data without violating SCM assumptions.

Based on counterfactual reasoning, many methods have been proposed to enhance the counterfactual fairness of models. The unawareness method requires that no sensitive feature be used in the decision-making process [16]. However, this method is limited by the correlation between other features and sensitive features, which can still lead to discrimination. To address this issue, Kusner et al. [5] suggested utilizing only those features in the causal graph that are not descendants of sensitive features during prediction. Although this approach can achieve complete counterfactual fairness, it significantly reduces the model's predictive performance. Currently, a widely used strategy to enhance counterfactual fairness involves adding a counterfactual fairness penalty term to the loss function during model training. Garg et al. [7] introduced a constraint on the difference in predicted outcomes between counterfactual pairs during training. CLAIRE learns fair representations through counterfactual fairness constraints and applies an invariance penalty to exclude variables associated with spurious correlations [8]. Zuo et al. [11] generated counterfactual samples using causal structures and combined these samples with the original ones using a symmetric function, creating CFRs. Su et al. [12] proposed MultiCFL, a framework designed to achieve multidimensional causal fairness by integrating both individual- and macro-level-sensitive features into decision models, and by constructing predictors based on the counterfactually fair descendants of sensitive features.

Although existing counterfactual fairness classification methods have made significant progress, they still exhibit several key limitations. First, most operate as closed boxes, providing limited insight into how fairness improvements are achieved. Second, they often rely on incomplete counterfactual reasoning frameworks, resulting in the generation of counterfactual samples that may violate the underlying causal relationships. Third, due to their lack of interpretability and transparency, current approaches often struggle to effectively balance classification performance with fairness. In contrast, CIRCUS is built upon a complete counterfactual reasoning framework and intuitively performs the causal intervention on selected features to generate CDSs. It then applies bias correction to the labels of these samples to generate counterfactually fair samples, thereby improving fairness while maintaining interpretability.

B. Causal Tabular Generative Adversarial Network

Datasets corresponding to causal models are typically tabular, which has sparked significant interest in tabular data synthesis. Unlike image data that only contains pixel values, tabular data encompasses diverse types such as categorical, discrete, and continuous variables, making its synthesis more challenging and demanding advanced data processing techniques. Table-GAN (TGAN) [17] is the first method to employ deep convolutional neural networks for general tabular data synthesis. To further address the difficulties of modeling both discrete and continuous columns, as well as mitigating class imbalance stemming from the original data distribution, Xu et al. [18] proposed conditional tabular GAN (CTGAN). Fourier CTGAN (FCT-GAN) [13] employs FNBs to capture global relations in tabular data, ensuring that the quality of the synthesized data remains stable regardless of the training column order.

As causal-aware learning gains prominence in data mining, many researchers are investigating the use of GANs to model causal relationships within datasets. DECAF [19] constructs generators with shared parameters to learn the observed causal conditions in the data. To integrate causal GANs with tabular GANs, Archer et al. [20] proposed a conditional semi-supervised GAN (CGAN) for general causal models that optimizes the corresponding causal mechanisms according to the causal structure and employs a maximum mean discrepancy (MMD) loss to ensure that the generated samples align with the empirical data distribution.

However, the aforementioned causal GAN methods replace exogenous variables in the SCM with random variables, without incorporating the inference process for exogenous variables. Moreover, these methods do not account for column invariance in the training datasets when constructing causal GANs and do not address the challenges posed by mixed variable types in tabular datasets. To overcome these limitations, our CITGAN uses an encoder to infer the exogenous variables and generators to generate causally consistent data, following the causal order. Additionally, CITGAN adopts the FNB module as the foundation for the generator of this architecture to handle complex data types in tabular datasets. Together, these components enable CITGAN to form a complete counterfactual reasoning framework, thereby supporting more accurate and causally consistent counterfactual data generation.

III. PRELIMINARIES

A. Notations

In this article, we use uppercase letters (e.g., X) to represent random features and lowercase letters (e.g., x) to denote specific values. The probability of X taking the value of x is expressed as $P(X = x)$. The variables X , S , U , and Y correspond to the nonsensitive features, sensitive features, unobserved exogenous variables, and the prediction label, respectively. Additionally, $\hat{Y}$ denotes the predicted label output by the predictor, and X_s represents the values of X when S is intervened to be s . Major notations are recorded in Table S(I) in the supplementary file.

B. Structural Causal Model

A commonly used formal framework for representing and inferring causal relationships is the SCM proposed by Pearl in [4], with the following relevant definitions.

Definition 1 (SCM): The SCM can be formally represented as a triple $M = (U, V, F)$ where

- 1) U is a set of unobserved variables determined by factors outside the model and cannot be caused by any variables in V , also known as exogenous variables.
- 2) V is a set of observed endogenous variables determined by the variables in $U \cup V$.
- 3) $F = \{f_1(\cdot), f_2(\cdot), \dots, f_{|V|}(\cdot)\}$ is a set of deterministic functions that describe causal relationships between variables, also known as structural equations. For each $V_i \in V$, the corresponding function f_i is a mapping from $U \cup (V \setminus \{V_i\})$ to V_i , that is, $V_i = f_i(Pa_{V_i}, U_{V_i})$, where $Pa_{V_i} \subseteq V \setminus \{V_i\}$ and $U_{V_i} \subseteq U$ represent the endogenous variables and the exogenous variables that directly determine V_i , respectively.

An SCM is often associated with a causal graph, which is represented as a directed acyclic graph (DAG). In a causal graph, causal relationships between nodes are represented by directed edges, which point from the parent variable to the child variable to indicate the direction of causality.

C. Intervention and Counterfactual Fairness

The Pearl Causal Hierarchy contains three levels: association, intervention, and counterfactuals. Intervention forcibly sets certain features to specific values, typically represented by the do-operator. Pearl [4] defines the do-operator as follows.

Definition 2 (Do-Operator): The do-operator for a variable V_i , denoted as $do(V_i = v)$, simulates a physical intervention that sets the value of V_i to v and removes all incoming edges to V_i in the causal graph. In an SCM M , this is implemented by replacing the structural equation $V_i = f_i(Pa_{V_i}, U_{V_i})$ with $V_i = v$.

The key distinction between interventions and counterfactuals lies in their objectives and conditioning contexts. An intervention alters the data-generating process by forcibly assigning a value to a variable, independent of its causes, and is typically employed to evaluate population-level effects under hypothetical scenarios. In contrast, counterfactual reasoning concerns hypothetical outcomes for a specific individual, grounded in the actual observations of that individual. Consider an individual with sensitive features $S = s$, nonsensitive features $X = x$, and an observed decision outcome $Y = y$. A typical counterfactual query is: "For this individual, what would $\hat{Y}$ have been if S had been set to s' ?" This query can be formally expressed as $P(\hat{Y}_{s'}(U) \mid X = x, S = s, Y = y)$ and computed in three steps: 1) *abduction*, which infers the posterior distribution of the exogenous variables U from their prior distribution given an individual (x, s, y) ; 2) *action*, which performs the intervention $S = s'$ by replacing the structural equation of S ; and 3) *prediction*, which computes the distribution of Y using the new structural equations together with the posterior of U . Building on the concept of counterfactual

reasoning, the definition of counterfactual fairness is presented as follows.

Definition 3 (Counterfactual Fairness [5]): A classifier $\hat{Y} = H(X, S)$ is said to satisfy counterfactual fairness under a given causal model if, for every instance (x, s) , it adheres to the following condition:

$$\begin{aligned} P(\hat{Y}_s(U) = y | X = x, S = s) \\ = P(\hat{Y}_{s'}(U) = y | X = x, S = s) \end{aligned} \quad (1)$$

for all y and $s' \in S \setminus \{s\}$, where $S \setminus \{s\}$ represents the set of all possible values of S excluding s and U represents the exogenous variables inferred from $P(U|X = x, Y = y, S = s)$ based on the causal model. Furthermore, $\hat{Y}_s(U) = H(X_s(U), s)$ denotes the predicted outcome of the classifier for counterfactual samples when the value of S is set to s .

D. Causal-GAN

Kocaoglu et al. [21] proposed causal-GAN, an extension of GANs designed to preserve the causal structure among features in a causal graph while generating both observational and interventional distributions. In this framework, the generator $G(Z)$ simulates a causal model to generate data, while the discriminator $D(G(Z))$ distinguishes between real and generated data. The noise variables Z correspond to the exogenous variables in the SCM and serve as the input to the generator. These noise variables can be decomposed as $\{Z_1, Z_2, \dots, Z_{|V|}\}$, where each Z_i corresponds to the exogenous variable U_i . Each structural equation f_i in the SCM is modeled by a separate generator $G_i : \mathbb{R}^{Pa(V_i)+1} \rightarrow \mathbb{R}$. Consequently, the generator $G(Z)$ can be decomposed into $|V|$ subnetworks $\{G_1, G_2, \dots, G_{|V|}\}$, with each subgenerator responsible for modeling a specific variable in accordance with the causal graph. The features are generated sequentially following the topological order of the causal DAG. If a node V_j is a parent node of V_i in the causal graph, the output of G_{V_j} is provided as an input to G_{V_i} , thereby ensuring that the generated data respects the causal relationships.

IV. CITGAN ARCHITECTURE

In this section, we provide the details of the CITGAN architecture, designed to model the underlying SCM of tabular datasets. The CITGAN architecture operates in three logical stages: 1) inferring exogenous variables via a regularized variational encoder; 2) generating data through a causal sequential generator; and 3) jointly optimizing the entire architecture via a multiobjective loss function.

A. Stage 1: Exogenous Variable Inference

In SCMs, exogenous variables U are typically present, and inferring these variables plays a crucial role in counterfactual reasoning. When the structural equations in the SCM are of the additive error form, that is, $V_i = f_i(Pa_i) + U_i$, inferring the value of the exogenous variable U_i from the data is relatively straightforward. However, when the structural equations are nonadditive, nonlinear transformation functions become necessary to infer these exogenous variables. To

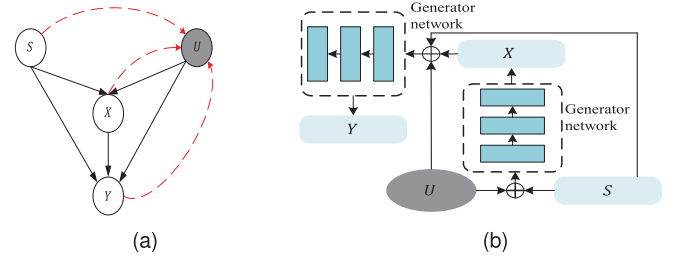


Fig. 1. Inference process of exogenous variables and corresponding VAE structure. (a) Inference process of exogenous variables. (b) Causal data generation.

address this challenge, we leverage a regularized variational encoder to map the observed sample space $\{X, S, Y\}$ into a latent exogenous embedding space U . As shown in Fig. 1(a), the encoder approximates the posterior distribution

$$u \sim \text{Enc}(x, s, y) = q(u|x, s, y). \quad (2)$$

To regularize this latent space, we enforce a prior distribution constraint $\mathcal{L}_{pri}$, assuming U follows a standard normal distribution $p(U) \sim N(0, 1)$:

$$\mathcal{L}_{pri} = \text{KL}[q(U|X, S, Y) \| p(U)] \quad (3)$$

where $\text{KL}[\cdot \| \cdot]$ denotes the Kullback–Leibler (KL) divergence between two distributions.

Fig. 1(a) illustrates the inference process of the exogenous variables, with red dashed lines representing posterior inference and black solid lines indicating prior inference. Given that there is no causal relationship between S and U , the inference process must ensure their independence. To accomplish this, we utilize two interconnected neural networks, GU and GS , to compute the Hirschfeld–Gebelein–Rényi (HGR) coefficient [22] between U and S , assessing their correlation and imposing the HGR correlation constraint, which is formulated as

$$\mathcal{L}_{hgr} = \max_{\omega_u, \omega_s} \mathbb{E}_{U \sim p(U), S \sim p(S)} \left(G\widehat{U}_{\omega_u}(U) G\widehat{S}_{\omega_s}(S) \right) \quad (4)$$

where ω_u and ω_s are parameters of network GU and network GS , respectively.

B. Stage 2: Causal Tabular Data Generation

Since CausalGAN can be adapted to generate data based on the topological structure of the causal graph, ensuring alignment with the causal relationships between endogenous variables, we employ CausalGAN to simulate the process of causal data generation. Feedforward neural networks are well-suited to model structural equations, as they can simulate nonlinear functions. Taking Fig. 1(a) as an example, the corresponding generator network structure is shown in Fig. 1(b), where the two generator networks represent the structural equations for X and Y , respectively. Each generator network is represented by a feedforward neural network, which generates data for the corresponding feature, with inputs including the parent variables and exogenous variables of that feature in the causal graph. Following the topological order of the causal DAG, each endogenous variable is modeled by its respective generator, as outlined below

$$\tilde{X}_i = G_i(\tilde{Pa}_i, U_i) \quad (5)$$

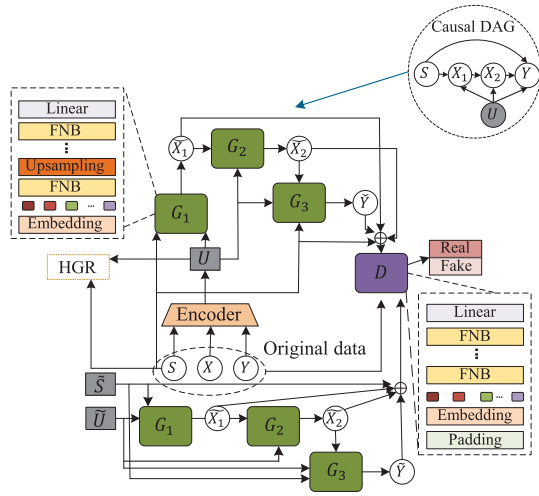


Fig. 2. Example of the CITGAN architecture, with the corresponding causal graph in the top-right corner. X represents nonsensitive features, including X_1 and X_2 .

where $\tilde{Pa}_i$ represents the causal parent node data of X_i generated by G_{Pa_i} . The causal sequential generator is denoted as $G(S, U) = [G_1(U), G_2(S, U), \dots, G_{|V|}(\cdot, U)]$, which generates data that aligns with the causal relationships of the corresponding SCM. The data generated in accordance with the causal order is then input into the discriminator D , which distinguishes between generated and real samples.

Given that tabular datasets have a unique structure, consisting of a mixture of categorical, discrete, continuous, and mixed values, we introduce several preprocessing steps [23] to encode data columns according to their respective types. Furthermore, we employ the multilayer FNB module [13] as the generator architecture, substituting the conventional network to better integrate global dependencies. This approach allows us to generate high-quality synthetic data while preserving the causal relationships among endogenous variables. The FNB module can improve the ability of the generated tabular data to simulate real-world scenarios and mitigate the impact of column order during training. Each FNB module consists of two main components: the Fourier layer and the feedforward network. The Fourier layer operates in three steps: 1) performing a 2-D discrete Fourier transform; 2) performing element-wise multiplication of the frequency-domain features with learnable weights; and 3) applying a 2-D inverse discrete Fourier transform.

C. Stage 3: Joint Training and Optimization

In the CITGAN architecture, as illustrated in Fig. 2, we integrate the inference encoder and the causal sequential generator into a unified end-to-end framework. The original data $\{X, S, Y\}$ is first mapped by the encoder network Enc into a latent space of exogenous variables U , which capture the unobserved factors independent of the sensitive features S . According to [24], the inference for the exogenous variables U can be modeled as

$$q_\phi(U|X, S, Y) = N(\mu_\phi(X, S, Y), \text{diag}(\sigma_\phi^2(X, S, Y))) \quad (6)$$

where $\mu_\phi(X, S, Y)$ and $\text{diag}(\sigma_\phi^2(X, S, Y))$ represent the mean and variance of the approximate posterior over U . And the exogenous variables U are then sampled via the reparameterization trick

$$U = Enc(X, S, Y) \\ = \mu_\phi(X, S, Y) + \sigma_\phi(X, S, Y) \odot \epsilon, \quad \epsilon \sim N(0, 1) \quad (7)$$

where $\odot$ denotes element-wise multiplication, ϵ is standard Gaussian noise, and ϕ represents the trainable parameters of the Enc . The encoder Enc is trained to minimize the HGR maximal correlation estimator between U and S , thereby ensuring that U is as independent as possible from S . The causal sequential generator $G(\cdot)$ utilizes S and U to generate reconstructed data. To strictly enforce causal consistency, CITGAN employs a topological generation process that operates sequentially according to the causal graph (e.g., $S \rightarrow X_1 \rightarrow X_2 \rightarrow Y$). Specifically, the process initiates by feeding the sensitive variable S and the corresponding exogenous variable U into the first subgenerator G_1 to produce $\tilde{X}_1$. Subsequently, the generated $\tilde{X}_1$ is fed into G_2 along with U to produce $\tilde{X}_2$, thereby structurally preserving the functional dependencies $X_1 = f(S, U)$ and $X_2 = f(X_1, U)$. Consequently, the final set of reconstructed data is obtained as $\{\tilde{X}, \tilde{S}, \tilde{Y}\} = G(S, U)$. Simultaneously, $\tilde{U}$ is sampled from its prior distribution $p(U)$ (a standard normal distribution) and input into the causal sequential generator, along with $\tilde{S}$ sampled from the distribution of S , to generate synthetic data: $\{\tilde{X}, \tilde{S}, \tilde{Y}\} = G(\tilde{S}, \tilde{U})$. Finally, the discriminator D judges whether the reconstructed data, synthetic data, and original data are real or fake.

The dashed boxes in Fig. 2 illustrate the network architectures. The generator begins by feeding the input vector into an embedding layer, which transforms and reshapes it into a feature map. This feature map is then processed through the FNB module, with an upsampling module applied after each FNB module to further refine the feature map. After passing through multiple FNB and upsampling modules, the embeddings are finally projected into a 1-D vector through a linear layer, representing the values of each feature. On the other hand, the discriminator receives a vector formed by concatenating all variables. This vector is first reshaped into an image using a padding layer, followed by convolutional filtering with an embedding layer to extract features. After passing through multiple FNB modules, the extracted features are flattened and reduced to a single value by the final linear layer, representing the discriminator's judgment of the input. The primary objective of the encoder and the causal sequential generator is to fit the implicit causal models within the dataset while ensuring that the distributions of reconstructed samples and synthetic samples closely match those of the real samples, thereby effectively deceiving the discriminator. Therefore, the encoder, generator, and discriminator can be jointly trained within a unified framework. To enhance training convergence and optimization stability, we adopt the Wasserstein GAN with gradient penalty (WGAN-GP) strategy [25]. By replacing the Jensen-Shannon divergence with the Wasserstein distance (WD) and enforcing the 1-Lipschitz constraint via gradient penalty, WGAN-GP provides informative gradients throughout the parameter space, effectively mitigating the vanishing

gradient problem. Accordingly, the CITGAN architecture is optimized using the following objective function:

$$\min_{\{G_i\}_{i=1}^{|V|}, \text{Encoder}} \max_D \mathcal{L} = \lambda_{pri} \mathcal{L}_{pri} + \lambda_{hgr} \mathcal{L}_{hgr} + \lambda_{rec} \mathcal{L}_{rec} + \lambda_{adv} \mathcal{L}_{GAN}. \quad (8)$$

Unlike standard GANs, we employ a dynamic weighting scheme where λ_{rec} and λ_{pri} decrease during training to prioritize representation learning and latent regularization in the early stages, while λ_{adv} increases to enhance adversarial refinement in later stages. The adversarial loss $\mathcal{L}_{GAN}$ utilizes the WD metric and is defined as

$$\begin{aligned} \mathcal{L}_{GAN} = & \mathbb{E}_{(x,s,y) \sim P_{data}(X,S,Y)} [D(x,s,y)] \\ & - \mathbb{E}_{(\tilde{x},s,\tilde{y}) \sim P_{G(S,U)}(X,S,Y)} [D(\tilde{x},s,\tilde{y})] \\ & - \mathbb{E}_{(\tilde{x},\tilde{s},\tilde{y}) \sim P_{G(\tilde{S},\tilde{U})}(\tilde{s},\tilde{s},\tilde{y})} [D(\tilde{x},\tilde{s},\tilde{y})] - \lambda_{gp} \mathcal{L}_{gp} \end{aligned} \quad (9)$$

where $\mathcal{L}_{gp}$ represents the gradient penalty term enforcing the 1-Lipschitz constraint. By removing logarithmic terms and incorporating gradient penalties, this formulation ensures stable gradient flow and prevents mode collapse. Furthermore, the training stability and convergence of the CITGAN architecture are empirically demonstrated in Fig. S.2 of the supplementary file. It is evident from the results that the model achieves robust convergence, effectively learning the underlying data distribution without suffering from mode collapse.

By optimizing (8), CITGAN is capable of both effectively capturing causal relationships and generating high-quality tabular data. The training process of CITGAN is outlined in Algorithm 1. The first line initializes the generator network for each feature and forms a causal sequential generator based on the causal order. The second line initializes the parameters for all components. Lines 5–19 outline the specific training procedure based on the objective function. Specifically, lines 5–9 outline the process of training the discriminator parameters θ_{Dis} using gradient ascent. Lines 10–19 describe the update of the generator parameters θ_{Gen} and the encoder parameters θ_{Enc} , which are optimized through K iterations of gradient descent. Within each of the K iterations, the parameters ω_u and ω_s of GU and GS are updated for H steps using gradient ascent to maximize the HGR estimate between U and S .

V. CIRCUS FRAMEWORK

In this section, we detail the CIRCUS framework for enhancing the counterfactual fairness of the trained classifier. The CIRCUS framework generates CDSs in two stages. First, in Section V-A, we generate CDSs near the decision boundary by performing causal interventions on the selected features of the original samples. Next, in Section V-B, we apply subtle causal interventions to the selected features of the previously generated discriminatory samples to further generate a larger number of discriminatory samples. Finally, in Section V-C, we employ bias correction preprocessing to the labels of these samples and retrain the classifier.

Algorithm 1 Training Process of CITGAN Architecture

Input: Instances of observable variables $\{X, S, Y\}$, training epochs T_{max} , causal graph $\mathcal{G}$

Output: $\theta_{Enc}, \theta_{Gen}, \theta_{Dis}, \omega_u, \omega_s$

- 1 Construct causal networks based on $\mathcal{G}$;
- 2 Initialize all network parameters;
- 3 Initialize epoch counter $t = 0$;
- 4 **repeat**
- 5 Randomly sample mini-batches $\{X, S, Y\}$;
- 6 Use the encoder to infer exogenous variables:
 $U = Enc(X, S, Y)$;
- 7 Sample $\tilde{U}$ and $\tilde{S}$;
- 8 Generate reconstructed samples and synthesized samples using $G(U, S)$ and $G(\tilde{U}, \tilde{S})$ respectively;
- 9 Update θ_{Dis} to maximize $\mathcal{L}_{GAN}$:
 $\theta_{Dis} \leftarrow \theta_{Dis} + \eta \nabla \mathcal{L}_{GAN}(\theta_{Dis})$;
- 10 **for** K steps **do**
- 11 Randomly sample mini-batches $\{X, S, Y\}$;
- 12 $U = Enc(X, S, Y)$;
- 13 **for** H steps **do**
- 14 Update ω_u and ω_s to maximize $\mathcal{L}_{hgr}$:
 $\omega_u \leftarrow \omega_u + \eta \nabla \mathcal{L}_{hgr}(\omega_u), \omega_s \leftarrow \omega_s + \eta \nabla \mathcal{L}_{hgr}(\omega_s)$;
- 15 $\mathcal{L}_{pri} = \text{KL}[q(U|X, S, Y)||p(U)]$;
- 16 Sample $\tilde{U}$ and $\tilde{S}$;
- 17 Generate reconstructed samples and synthesized samples using $G(U, S)$ and $G(\tilde{U}, \tilde{S})$ respectively;
- 18 $\mathcal{L}_{rec} = \mathbb{E}_{q(U|X,S,Y)} [-\log(p(G(U, S)|U, S))]$;
- 19 Use Eq. (9) to calculate $\mathcal{L}_{GAN}$ and update θ_{Gen} and θ_{Enc} to minimize $\mathcal{L}$:
 $\theta_{Gen} \leftarrow \theta_{Gen} - \eta \nabla \mathcal{L}(\theta_{Gen}), \theta_{Enc} \leftarrow \theta_{Enc} - \eta \nabla \mathcal{L}(\theta_{Enc})$;
- 20 $t = t + 1$;
- 21 **until** $t \geq T_{max}$;

A. Global Generation

The CIRCUS framework generates CDSs by applying causal interventions to selected features. The definition of CDSs is as follows.

Definition 4 (CDS): If the prediction results of counterfactual samples, generated by altering different values of sensitive features of a sample (x, s, y) , are inconsistent, the sample (x, s, y) is considered a CDS, which can be formulated as

$$\exists i, j (1 \leq i < j \leq k) \quad H(x_{s_i}(u), s_i) \neq H(x_{s_j}(u), s_j) \quad (10)$$

where u denotes the value of exogenous variable inferred from (x, y, s) using the CITGAN architecture, and k represents the number of unique values for the sensitive features.

Inspired by the robustness studies of deep neural networks (DNNs) [26] and previous work [27] that utilizes the gradient of the prediction loss to identify adversarial samples, we leverage the gradient of the classifier's prediction loss ($\nabla_x J(\theta, x, y)$) on counterfactually fair samples to guide the generation of CDSs during the global generation phase. In this section, our focus is on differentiable classifiers, such

as DNNs, whose decision boundaries are generally smooth, thereby ensuring that gradient-based perturbations can reliably indicate directions that increase counterfactual discrimination. As illustrated in Fig. 3, the direction of the prediction loss gradient with respect to the sample features indicates the direction required to cross the decision boundary, while the gradient magnitude reflects the difficulty of crossing the boundary, with a larger magnitude making the crossing easier. To maximize the likelihood of samples crossing the decision boundary, we focus on the features with the largest gradient magnitudes and perform causal interventions along the gradient direction. This strategy increases the probability of changing the predicted outcome of a sample, which in turn enhances the likelihood of altering the prediction of counterfactual samples, thus ensuring that the generated samples conform to the definition of the CDS.

During training, the classifier learns not only the relationship between features and prediction outcomes but also the dependency between sensitive and nonsensitive features. Due to this dependency, nonsensitive features may serve as proxies for sensitive features, transmitting sensitive information to the prediction outcomes through causal paths [28]. Intuitively, if an original sample is counterfactually fair, its prediction can be considered fair with respect to the sensitive features. However, intervening on nonsensitive features that are strongly associated with the sensitive attribute may indirectly alter the influence of the sensitive features on the prediction outcome. The degree of association can be assessed by examining whether a nonsensitive feature is a descendant of the sensitive features in the causal graph, with closer descendants indicating stronger associations. For example, when the sensitive feature S affects the prediction outcome Y through two causal paths (e.g., $S \rightarrow X_1 \rightarrow X_2 \rightarrow Y$, $S \rightarrow X_1 \rightarrow X_3 \rightarrow Y$), intervening on an early descendant such as X_1 can simultaneously block both paths, whereas intervening on later descendants such as X_2 or X_3 can block only a single path. Therefore, for counterfactually fair samples, interventions on nonsensitive features located earlier in the causal ordering are more likely to convert them into CDSs.

In the global stage, the selection of the intervention feature considers the gradient magnitude of nonsensitive features with respect to the sample's prediction loss and their association degree with sensitive features. The method for selecting the intervention feature I_x in the global generation for the sample x is formulated as

$$r(X_i, S) = \begin{cases} |X| - \text{rank}(X_i, \text{De}(S)), & \text{if } X_i \in \text{De}(S) \\ |X| - |\text{De}(S)|, & \text{otherwise} \end{cases} \quad (11)$$

$$I_x = \underset{i}{\operatorname{argmax}} (|\nabla_{x_i} J(\theta, x)| * r(X_i, S)) \quad (12)$$

where $\nabla_{x_i} J(\theta, x)$ is the gradient of the loss function $J(\theta, x)$ with respect to the feature x_i of the sample x , $|X|$ denotes the number of nonsensitive features, and $\text{De}(S)$ represents the set of descendant nodes of S obtained using the search algorithm. The term $\text{rank}(X_i, \text{De}(S))$ denotes the position of X_i in the topological order of the descendants of S based on the causal graph. I_x represents the most important feature to be intervened upon when generating a CDS. Intervening on I_x also causes

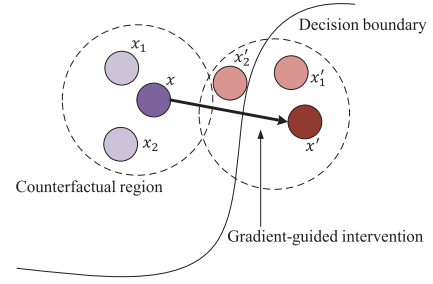


Fig. 3. Intuition of gradient-guided intervention.

Algorithm 2 Global Generation

Input: Instances of observable variables $\{X, S, Y\}$,

g_{num}, max_iter, g_s

Output: Global CDS set G_{cds}

```

1 Initialize  $G_{cds} = \emptyset$ ;
2 for  $i$  from 0 to  $g_{num}$  do
3   Get seed  $x$  from original dataset;
4   for  $iter$  from 0 to  $max\_iter$  do
5     if  $x$  is a CDS then
6        $G_{cds} = G_{cds} \cup \{x\}$ ;
7       break;
8     Compute  $grad = \nabla J(\theta, x)$  and determine
       feature  $I_x$ ;
9      $x^{(I_x)} = x(I_x) + g_s * \text{sign}(grad_{I_x})$ ;
10    Intervene the value of  $I_x$  to  $x^{(I_x)}$  using
       CITGAN to generate new  $x$ ;
```

its descendant features to change accordingly based on the SCM, thereby ensuring causal consistency in the generated samples. Within the CITGAN architecture, the intervention operation for sample x follows the definition of the do-operator by replacing the output of the generator corresponding to I_x with the specified intervention value $x^{(I_x)}$.

Algorithm 2 outlines the details of the global generation phase, where g_{num} represents the number of samples to be generated during this phase, max_iter denotes the maximum number of iterations, g_s is the step size for global generation, and G_{cds} is the set of CDSs generated during this phase. In each iteration, a seed instance x is sampled from the original dataset (line 3). The algorithm then checks whether x meets the conditions for being a CDS (line 5). If the condition is met, x is added to the set G_{cds} , and the iteration terminates (lines 6 and 7). If not, the gradient of the loss function is calculated, and the intervention feature is determined based on (11) and (12), and x is updated through causal intervention using the CITGAN architecture to generate a new sample (line 10). This process repeats until either a CDS is found or the maximum number of iterations is reached.

As the first step in generating counterfactually fair samples, the global generation stage lays the foundation for enhancing the counterfactual fairness of the classifier by generating CDSs, which are later transformed into counterfactually fair samples through label adjustment.

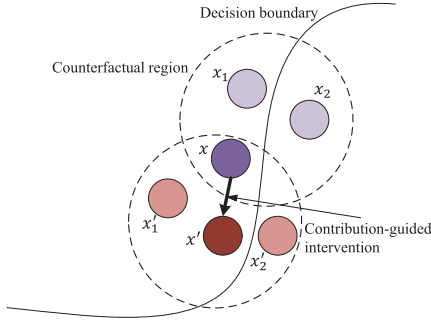


Fig. 4. Intuition of contribution-guided intervention.

B. Local Generation

Upon completing the global generation phase, we obtain a set of CDSs. However, because the number of CDSs generated is limited, it becomes necessary to generate more CDSs based on the generated discriminatory samples. Given the intuition that a well-trained classifier should exhibit robustness, it should produce the same prediction for similar samples. This suggests that numerous similar CDSs likely exist near the generated CDSs. Therefore, the local generation phase applies minimal causal interventions to these CDSs to generate additional ones.

As shown in Fig. 4, in the local generation stage, since the input samples are already counterfactually discriminatory, it is crucial to ensure that their prediction outcomes remain as stable as possible after causal intervention while preserving the influence of the sensitive features on the prediction. To achieve this, causal interventions in this stage should target nonsensitive features that contribute minimally to the prediction outcome and exhibit a low association degree with the sensitive features. To quantify feature contributions, a Shapley additive explanation (SHAP) interpreter is initialized for the trained classifier using the original dataset and the generated samples, which evaluates the contribution sv_{x_i} of each feature x_i in sample x to its prediction outcome. According to the properties of the Shapley value [29], features with smaller contributions are less likely to alter the prediction when intervened upon. Based on the feature's contribution to the prediction and its association degree with sensitive features, the normalized probability of each nonsensitive feature being selected as an intervention feature is defined as

$$p_{x_i} = \frac{\frac{1}{r(X_i, S)} \cdot \frac{1}{sv_{x_i}}}{\sum_{j=1}^d \frac{1}{r(X_j, S)} \cdot \frac{1}{sv_{x_j}}} \quad (13)$$

where d represents the number of features excluding the label.

Algorithm 3 outlines the steps of the local generation phase, where l_{num} represents the number of samples to be generated, l_s denotes the step size, and L_{cds} is the set of CDSs generated during the local generation phase. The process begins by initializing a SHAP explainer for the classifier using a mixed dataset (line 1). In line 3, we select a seed sample x from the set G_{cds} . Next, in lines 5–7, we calculate the probability of nonsensitive features being selected in x using Equation 13

Algorithm 3 Local Generation

Input: Instances of observable variables $\{X, S, Y\}$,

l_{num}, l_s, G_{cds}

Output: Local CDS set L_{cds}

```

1 Initialize SHAP explainer using mixed data;
2 Initialize  $L_{cds} = \emptyset$ ;
3 for  $x \in G_{cds}$  do
4   for  $k$  from 0 to  $l_{num}$  do
5     Calculate the contribution  $sv_{x_i}$  of  $x_i$  to
       prediction;
6     Calculate the intervention probability  $p_{x_i}$ ;
7     Select  $I_x \in X \setminus S$  with probability  $p_{I_x}$ ;
8     Select  $sign \in [-1, 1]$  with probability
        $[0.5, 0.5]$ ;
9      $x^{(I_x)} = x(I_x) + sign * l_s$ ;
10    Intervene the value of  $I_x$  to  $x^{(I_x)}$  using
       CITGAN to generate new  $x$ ;
11    if  $x$  is a CDS then
12       $L_{cds} = L_{cds} \cup \{x\}$ ;
```

and select the intervention feature. Line 8 involves choosing a random intervention direction (sign) with uniform probability. The CITGAN architecture is then employed to intervene on x to generate new samples (lines 9 and 10). Subsequently, we check whether the intervened sample remains a CDS (lines 11–12). This procedure continues until the number of newly generated samples reaches the predefined l_{num} . The CIRCUS framework integrates both global and local generation phases to generate CDSs, as depicted in Fig. 5. The global generation phase corresponds to the left dashed box in the figure, while the local generation phase corresponds to the right dashed box. The dashed arrows represent the construction of used components in CIRCUS. Both the global and local generation stages involve discrimination detection and causal intervention on a sample (x, s, y) . The discrimination detection process proceeds as follows: first, the sample (x, s, y) is fed into the encoder of the CITGAN framework to infer the values of the exogenous variables u ; next, the value of sensitive feature S is replaced with a different value s' , and counterfactual samples are generated following the causal order using the causal sequential generator as $G(s', u) = (x', s', y')$; finally, the prediction results $H(x, s)$ and $H(x', s')$ are compared. During the causal intervention process, new samples $G(s, u)$ are generated sequentially according to the causal order, based on the value of the inferred exogenous variable u and the value of sensitive feature s . For the intervened feature I_x , the intervention operation $do(I_x = x^{(I_x)})$ is applied, whereby the original generation process $I_x = G_{I_x}(Pa_{I_x}, u_{I_x})$ is replaced with $I_x = x^{(I_x)}$, and this replacement constitutes an integral part of the causal intervention.

C. Bias Correction Preprocessing

The classifier learns the distribution $p(y|x, s)$ from the underlying SCM in the original data. However, when a large number of CDSs are included in the training data, the classifier may inadvertently inherit such biases, resulting in counterfactual

TABLE I
DATASET DESCRIPTION

Dataset	Instance	Feature	Sensitive Feature	Task
Adult[30]	32,561	10	Sex	Predict whether an individual's annual income exceeds \$50K
Credit[30]	690	14	Ethnicity	Predict whether an applicant will be approved for a credit card
Bank[30]	45,211	14	Age	Predict whether a customer will subscribe to a term deposit
Bail[31]	20,856	15	White	Predict whether a defendant is high risk of recidivism
Nutrition[32]	9,932	18	Race	Predict whether an individual survives more than 15 years

discrimination. To improve counterfactual fairness, we apply bias correction preprocessing to the labels of generated samples after obtaining CDSs, ensuring their compliance with the counterfactual fairness constraint as shown below

$$\forall i, j (1 \leq i < j \leq k) \quad y_{s_i} = y_{s_j} = y \quad (14)$$

where y_{s_i} is the label of the generated counterfactual sample (x_{s_i}, s_i, y_{s_i}) when the sensitive feature S takes the value s_i .

When generating (x_{s_i}, s_i, y_{s_i}) , similar to the procedure described above, the value of the exogenous variable u is first inferred, after which the value of the sensitive feature is directly replaced with s_i , and the counterfactual sample $G(s_i, u)$ is generated using the causal sequential generator. The labels of all counterfactual samples corresponding to the generated CDS (x, s, y) are then replaced with the label feature y from the generated CDS. These corrected samples are subsequently combined with the original training data to retrain the classifier. Since the generated samples adhere to the causal relationships, retraining on samples that have undergone bias correction preprocessing can improve the counterfactual fairness of the classifier. Moreover, adjusting the labels of counterfactual samples in the generated dataset does not alter the feature distribution of the original data. Consequently, retraining the classifier with label-adjusted samples has minimal impact on its classification performance on the original data.

VI. EXPERIMENTS

In this section, we first evaluate the quality of the data generated by the CITGAN architecture, focusing on its consistency with the statistical properties of the original data and the preservation of causal relationships. Second, we conduct an ablation study on the intervention feature selection strategy within the CIRCUS framework to demonstrate the effectiveness of CIRCUS in generating CDSs. Third, we evaluate the effectiveness of the proposed CIRCUS framework in improving the counterfactual fairness of the classifier by comparing CIRCUS with several state-of-the-art methods. The experimental setup, including the datasets and evaluation metrics used, is introduced below.

A. Experimental Settings

Datasets: The experimental section uses five real-world datasets that are widely adopted in fairness research: adult census income [30], credit card approval [30], bank marketing [30], Bail [31], and nutrition [32]. Table I summarizes the statistical information, sensitive feature, and task for each dataset.

We apply the causal graphs from [19] to both the Adult dataset and the Credit dataset and use the causal graph from [33] for the Nutrition dataset. For the Bank and Bail datasets, we employ Tetrad [34] to generate the corresponding causal graphs using the Boss algorithm [35]. To ensure reproducibility, supporting details are provided in the supplementary file: Part I presents the dataset descriptions, while Fig. S.1 illustrates the causal graphs for the Bank and Bail datasets. Furthermore, Part II outlines the implementation details, with the specific hyperparameters listed in Table S(II).

Evaluation metrics for evaluating data generation methods: ML utility and statistical similarity.

- 1) *ML Utility:* We train three classifiers—DNN, logistic regression (LR), and residual network (ResNet)—on original and synthetic data, respectively, and evaluate their performance on the same test set. We assess the models using accuracy (ACC) and area under the curve (AUC), quantifying the performance differences between models trained on the original and synthetic data with ACC difference (ACD) and AUC difference (AUD). Smaller values indicate that synthetic data approaches the quality of the original data. The final results are averaged across all three classifiers.
- 2) *Statistical Similarity:* We evaluate the statistical similarity between real and synthetic data using Average Jensen–Shannon divergence (JSD), average WD, and difference in pair-wise correlation (Corr). JSD measures differences in the distributions of categorical variables, WD assesses the distributions of continuous variables, and Corr evaluates the preservation of column dependencies. Smaller values of JSD and WD indicate higher similarity, while a smaller Corr value reflects superior modeling of global relations.

Evaluation metrics for evaluating methods to improve counterfactual fairness: classifier performance and counterfactual fairness.

- 1) *Classifier Performance:* We assess performance using ACC and the $F1$ -score. A higher value of these metrics indicates better classification performance.
- 2) *Counterfactual Fairness:* We follow [8] and [36] in using MMD to measure distributional divergence for evaluating counterfactual fairness. Specifically, we quantify the discrepancy between the prediction distributions of each pair of counterfactual samples ($S \leftarrow s$ and $S \leftarrow s' \forall s \neq s'$) generated from the same original sample, and then take the average as the final result. We employ two variants of MMD: linear MMD (MMD_L)

TABLE II
COMPARING DIFFERENT DATA GENERATION APPROACHES

Criteria	FCT-GAN	DECAF	CGAN	VAEs	CITGAN
C1	×	×	×	✓	✓
C2	×	✓	✓	×	✓
C3	×	✓	✓	×	✓
C4	×	×	×	✓	✓
C5	✓	×	×	×	✓

and Kernel-based MMD (MMD_K). Smaller MMD values indicate higher counterfactual fairness.

B. Evaluation of Generated Data Quality

We compare CITGAN with seven data synthesizers, including CVAE [11], DCEVAE [11], CLAIRE-VAE [8], DECAF [19], FCT-GAN [13], MultiCFL-VAE [12], and CGAN [20], and conduct experiments on the five datasets described earlier.

To highlight the distinctions between the baselines and our approach, we summarize their functional differences in Table II. We differentiate these methods across five criteria: C1) whether exogenous variables can be inferred; C2) whether causal relationships between endogenous variables can be preserved; C3) whether intervention operations can be performed; C4) whether counterfactual samples can be generated; and C5) whether the method is suitable for complex tabular data types. CVAE, DCEVAE, CLAIRE-VAE, and MultiCFL-VAE are categorized into the same group, as they all employ a VAE encoder to infer exogenous variables from the data and subsequently use a decoder to generate counterfactual samples by combining the exogenous variables with sensitive attributes, thereby satisfying the first and fourth criteria. However, VAEs cannot model the causal relationships between endogenous variables and therefore do not satisfy the second and third criteria. DECAF and CGAN learn the causal relationships in the data and begin inference from the root nodes of the causal graph. They synthesize features in topological order, thereby preserving causal relationships between features and enabling interventions during the data generation process. However, they cannot infer exogenous variables, and consequently cannot generate counterfactual samples. Among the evaluated frameworks, only CITGAN and FCT-GAN are explicitly designed for tabular data, with both utilizing the FNB module as the generator backbone. However, FCT-GAN employs a single generator to model all features simultaneously without causal constraints, thereby failing to meet the first four criteria. In contrast, CITGAN assigns a dedicated subgenerator to each feature, generating values sequentially following the topological order. This design strictly preserves causal dependencies and ensures SCM fidelity. Consequently, by integrating exogenous inference with causal generation, CITGAN uniquely fulfills all five criteria.

We conduct ten experiments on each dataset and compute the average of the metrics. Table III presents the ML utility and statistical similarity across eight data synthesizers. The results indicate that VAEs, DECAF, and CGAN exhibit mediocre performance across all metrics, showing only modest success in a few cases. This indicates that they are limited in modeling the global distribution and capturing global dependencies,

TABLE III
COMPARISON OF DATA QUALITY GENERATED BASED ON REAL DATASETS

Dataset	Synthesizer	ML Utility		Statistical Similarity		
		ACD	AUD	JSD	WD	Corr
Adult	CVAE	0.162	0.173	0.233	0.066	1.599
	DCEVAE	0.061	0.134	0.250	0.069	2.228
	CLAIRE-VAE	0.042	0.073	0.167	0.061	1.451
	MultiCFL-VAE	0.055	0.146	0.218	0.103	2.816
	DECAF	0.117	0.304	0.782	0.042	3.767
	FCT-GAN	0.019	0.032	0.094	0.027	0.750
	CGAN	0.051	0.301	0.493	0.024	2.705
	CITGAN	0.007	0.011	0.023	0.024	0.394
Credit	CVAE	0.139	0.178	0.123	0.081	5.296
	DCEVAE	0.147	0.219	0.160	0.062	5.129
	CLAIRE-VAE	0.150	0.165	0.275	0.065	6.629
	MultiCFL-VAE	0.097	0.146	0.284	0.062	3.479
	DECAF	0.218	0.411	0.674	0.498	5.439
	FCT-GAN	0.020	0.031	0.054	0.014	2.287
	CGAN	0.239	0.166	0.433	0.056	4.997
	CITGAN	0.000	0.008	0.012	<u>0.054</u>	1.782
Bank	CVAE	0.095	0.049	0.018	0.062	1.784
	DCEVAE	0.015	0.075	0.010	0.062	1.204
	CLAIRE-VAE	0.021	0.082	0.009	0.076	1.413
	MultiCFL-VAE	0.047	0.241	0.130	0.059	2.121
	DECAF	0.016	0.377	0.391	0.071	4.240
	FCT-GAN	0.007	0.064	0.088	0.017	0.891
	CGAN	0.016	0.249	0.415	0.100	3.742
	CITGAN	0.001	0.015	0.009	<u>0.041</u>	0.224
Bail	CVAE	0.014	0.009	0.014	0.021	0.649
	DCEVAE	0.023	0.061	0.004	0.023	0.716
	CLAIRE-VAE	0.008	0.026	0.007	0.023	0.675
	MultiCFL-VAE	0.020	0.041	0.275	0.032	1.214
	DECAF	0.114	0.160	0.568	0.039	4.153
	FCT-GAN	0.026	0.051	0.033	0.011	1.230
	CGAN	0.160	0.108	0.390	0.116	2.883
	CITGAN	0.005	0.008	0.003	0.021	0.257
Nutrition	CVAE	0.058	0.210	0.285	0.063	13.347
	DCEVAE	0.064	0.026	0.006	0.066	10.100
	CLAIRE-VAE	0.059	0.092	0.254	0.066	10.854
	MultiCFL-VAE	0.068	0.116	0.017	0.113	4.821
	DECAF	0.056	0.278	0.506	0.359	14.396
	FCT-GAN	0.011	0.022	0.018	0.012	1.612
	CGAN	0.042	0.112	0.282	0.090	7.217
	CITGAN	0.056	0.022	0.005	0.023	2.247
Average	CVAE	0.094	0.124	0.135	0.059	4.535
	DCEVAE	0.062	0.103	0.086	0.056	3.875
	CLAIRE-VAE	0.056	0.088	0.142	0.058	4.204
	MultiCFL-VAE	0.057	0.138	0.185	0.074	2.890
	DECAF	0.104	0.306	0.584	0.202	6.399
	FCT-GAN	0.017	0.040	0.057	0.016	1.354
	CGAN	0.102	0.187	0.403	0.077	4.309
	CITGAN	0.014	0.013	0.010	0.033	0.981

The best results are shown in bold, and the runner-up results are underlined. The "Average" row represents the mean of the experimental results from all datasets.

primarily because they cannot construct a complete causal model. In addition, their inability to effectively adapt to the characteristics of tabular data hinders them from fully learning the complex feature structures inherent in such datasets. In contrast, FCT-GAN performs well on the Nutrition, Adult, and Credit datasets, showcasing its effectiveness in learning complex characteristics of tabular data. However, due to its lack of modeling the intrinsic causal relationships within the data, its performance remains inferior to that of our proposed CITGAN. The proposed CITGAN architecture stands out, achieving outstanding performance on ML utility metrics. Specifically, CITGAN outperforms all baselines on the AUD metric across all datasets, with an average ACD value of 17.6% and an average AUD value of 67.5% better than the

TABLE IV
EXPERIMENTAL RESULTS OF THE ABLATION STUDY ON INTERVENTION FEATURE SELECTION

Method	Adult			Credit			Bail			Nutrition		
	$ G_{cds} $	$ L_{cds} $	$\frac{ L_{cds} }{ G_{cds} }$	$ G_{cds} $	$ L_{cds} $	$\frac{ L_{cds} }{ G_{cds} }$	$ G_{cds} $	$ L_{cds} $	$\frac{ L_{cds} }{ G_{cds} }$	$ G_{cds} $	$ L_{cds} $	$\frac{ L_{cds} }{ G_{cds} }$
Random	53	1096	20.68	49	758	15.47	76	2080	27.37	66	3011	45.62
Non-Association	61	1873	30.70	55	1425	25.91	84	3043	36.23	78	4111	52.71
CIRCUS	69	2543	36.86	60	1817	30.28	89	3720	41.80	86	5372	62.47

second-best. These results suggest that the data generated by CITGAN more closely approximates the distribution of real data, enabling classifiers trained on synthetic data to achieve test performance comparable to those trained on real data.

Regarding statistical similarity metrics, CITGAN significantly outperforms all baseline methods on the JSD metric. Specifically, on the Adult dataset, CITGAN achieves a JSD score 75.5% better than the runner-up method, and on the Credit dataset, it is 77.8% better. These results highlight CITGAN's superior performance in modeling categorical feature distributions. In terms of the WD metric, CITGAN performs comparably to CGAN on the Adult dataset and ranks among the best, while on other datasets, its performance is slightly lower than that of the top-performing method, FCT-GAN, ranking second overall, while still surpassing all other synthesizers with causal modeling capability. This is because, when continuous variables are generated in causal order, generation deviations in earlier variables may gradually accumulate and affect subsequent nodes. In contrast, FCT-GAN lacks causal consistency and relies solely on intercolumn correlations to generate variables. Nevertheless, in the joint evaluation of JSD and WD, CITGAN achieves the best results across all datasets, highlighting its strong ability to capture feature distributions and comprehensively reflect the distributional characteristics of the source data. In the Corr metric evaluation, CITGAN significantly outperforms the baseline algorithms on most datasets. Notably, CITGAN performs 60.4% better than the second-best method on the Bail dataset and 74.9% better on the Bank dataset. On average, CITGAN outperforms all other methods, demonstrating its effectiveness in generating causal data and modeling global relationships. Furthermore, the Wilcoxon signed-rank test [37] was conducted to assess statistical significance, and the results are listed in Tables S(III)–S(VI) of the supplementary file, with p -values indicating significance. These results statistically confirm that the performance improvements over the baselines are significant rather than due to random chance.

The experimental results demonstrate that CITGAN can generate high-quality tabular data while preserving causal relationships between features. Furthermore, CITGAN excels in modeling global relationships, which is closely related to the goal of capturing causal relationships between features. These advantages make CITGAN a powerful framework for advancing counterfactual reasoning and enhancing counterfactual fairness.

C. Ablation on FNB Module in CITGAN Architecture

To evaluate the specific contribution of the FNB module, we conduct an ablation study comparing the full model

TABLE V
ABLATION STUDY ON NETWORK ARCHITECTURE COMPARING CITGAN WITH CITGAN-MLP

Dataset	Method	ACD	AUD	JSD	WD	Corr
Adult	CITGAN-MLP	0.008	0.012	0.033	0.045	0.576
	CITGAN	0.007	0.011	0.023	0.024	0.394
Credit	CITGAN-MLP	0.048	0.020	0.026	0.055	1.866
	CITGAN	0.000	0.008	0.012	0.054	1.782
Bank	CITGAN-MLP	0.009	0.086	0.062	0.043	0.886
	CITGAN	0.001	0.015	0.009	0.041	0.224
Bail	CITGAN-MLP	0.011	0.022	0.005	0.033	0.295
	CITGAN	0.005	0.008	0.003	0.021	0.257
Nutrition	CITGAN-MLP	0.065	0.107	0.015	0.064	4.264
	CITGAN	0.056	0.022	0.005	0.023	2.247

with a simplified variant, CITGAN-MLP, where the specialized FNB module is replaced by standard MLPs. This configuration deliberately omits the Fourier expansion and frequency-domain modeling components featured in the full CITGAN model. By adopting this streamlined MLP-based structure, we aim to isolate the performance gains attributable to the causal DAG framework from those originating from the FNB module. The comparative results are summarized in Table V.

Notably, even the simplified CITGAN-MLP variant demonstrates competitive performance when compared to other state-of-the-art causal generative models listed in Table III, such as DECAF and MultiCFL-VAE. For instance, on the Adult dataset, CITGAN-MLP achieves an ACD of 0.008, significantly outperforming CGAN (0.051) and MultiCFL-VAE (0.055). This baseline strength stems from the CITGAN framework itself, which strictly enforces causal consistency through the topological generation process described in Section IV-C, ensuring that synthetic data respects the underlying cause-effect relationships. However, the full CITGAN significantly enhances this performance by incorporating the FNB module. While the causal framework ensures structural correctness, standard MLPs struggle to capture complex, non-local dependencies. The FNB module addresses this limitation by leveraging spectral transformations to model global correlations effectively. This impact is particularly evident in the Bank dataset, where the Corr value decreases substantially from 0.886 to 0.224. The robustness of these improvements is further validated by the violin plots across all datasets, presented in Figs. S.3–S.7 of the supplementary file.

D. Ablation on Intervention Feature Selection in CIRCUS

Since CIRCUS requires a sufficient number of CDSs to enhance counterfactual fairness, the number of generated samples largely depends on the strategy for selecting intervention features. To evaluate the effectiveness of CIRCUS in this

regard, we conduct an ablation study and compare it with two baselines: random (randomly selecting intervention features) and nonassociation (selecting intervention features strictly based on gradient magnitudes in the global stage, analogous to adversarial attack mechanisms, and based on feature contributions in the local stage). In the experiments, we set both g_{num} and l_{num} to 100, and use a trained DNN classifier to compare the number of CDSs generated in the global and local stages, denoted as $|G_{cds}|$ and $|L_{cds}|$, along with their ratio $|L_{cds}|/|G_{cds}|$. This ratio reflects the average number of additional samples generated in the local stage for each CDS from the global generation phase. As shown in Table IV, the nonassociation strategy generates more CDSs than the random strategy in both stages, confirming that guiding intervention via gradients (analogous to adversarial attacks) and feature contributions is inherently superior to blind selection. However, CIRCUS significantly outperforms the nonassociation strategy in both stages, validating the necessity of incorporating association degrees. While gradient-based adversarial attacks seek the path of least resistance to flip predictions, they do not guarantee the discovery of valid CDSs. As discussed in Section V-A, effective fairness intervention requires targeting early descendants (e.g., X_1 in the chain $S \rightarrow X_1 \rightarrow \{X_2, X_3\} \rightarrow Y$) to block the causal transmission of sensitive information. By prioritizing such high-association features, CIRCUS uncovers deep-seated discrimination rather than mere adversarial vulnerabilities. This leads to a significantly higher yield of CDSs in the global stage ($|G_{cds}|$) and ensures that the local search ($|L_{cds}|$) remains focused on discriminatory subspaces rather than diverging into irrelevant predictive features.

E. Evaluation on Improving Counterfactual Fairness

We compare the CIRCUS framework with seven baselines: unfair (UF), which trains a supervised model using all observable features without any fairness constraint, unawareness (unaware) [16], counterfactual training without descendants (CD) [5], counterfactual regularizer (CR) [7], CFR [11], counterfactually fair and invariant predictor (CLAIRE) [8], and MultiCFL [12].

Due to the large number of samples and sensitive feature values in the Bank dataset, applying it in this experiment poses significant challenges. Therefore, we select the remaining four datasets for the comparison experiment. We assess the effectiveness of various methods in improving the counterfactual fairness of both the traditional DNN classifier and the more complex ResNet classifier. Additionally, we split the training/validation/test set for all these four datasets as 60%/20%/20%. To evaluate fairness metrics, we use CITGAN to generate counterfactual samples based on the test set and calculate MMD_L and MMD_K . The CR method requires setting the weight parameter for the fairness constraint loss term, and the CLAIRE method requires setting the weight parameters for both the counterfactual fairness constraint and invariance loss terms. To find weight parameters that can maintain good classification performance while achieving a balance in fairness metrics, we explore values within the range of 0–10. The implementation of CIRCUS requires configuring g_{num} and l_{num} . In practice, parameter selection balances computational

TABLE VI
EXPERIMENTAL RESULTS OF THE COUNTERFACTUAL FAIRNESS ENHANCEMENT METHOD FOR THE DNN

Dataset	Method	Metric			
		ACC	F1-score	MMD_L	MMD_K
Adult	UF	0.824	0.618	2626.127	0.492
	Unaware	0.823	0.616	1645.094	0.302
	CD	0.746	0.000	-	-
	CR	0.755	0.176	659.198	0.262
	CFR	0.820	0.612	1160.568	0.237
	CLAIRE	0.753	0.159	362.091	0.160
	MultiCFL	0.781	0.513	334.453	0.198
	CIRCUS	0.810	0.623	234.865	0.116
Credit	UF	0.866	0.870	106.430	0.015
	Unaware	0.836	0.850	0.107	0.000
	CD	0.806	0.806	-	-
	CR	0.858	0.859	14.738	0.005
	CFR	0.828	0.839	2.571	0.000
	CLAIRE	0.843	0.847	8.651	0.003
	MultiCFL	0.843	0.858	1.115	0.001
	CIRCUS	0.873	0.879	0.191	0.000
Bail	UF	0.660	0.472	451.922	0.063
	Unaware	0.647	0.457	10.738	0.002
	CD	0.641	0.144	-	-
	CR	0.659	0.377	35.588	0.027
	CFR	0.653	0.466	7.616	0.006
	CLAIRE	0.651	0.469	6.027	0.005
	MultiCFL	0.643	0.470	35.499	0.035
	CIRCUS	0.651	0.497	4.127	0.002
Nutrition	UF	0.725	0.812	1112.568	0.118
	Unaware	0.723	0.808	433.482	0.081
	CD	0.697	0.778	-	-
	CR	0.717	0.802	62.049	0.073
	CFR	0.717	0.771	28.027	0.010
	CLAIRE	0.711	0.799	52.930	0.038
	MultiCFL	0.724	0.812	51.479	0.010
	CIRCUS	0.737	0.822	15.777	0.006
Average	UF	0.769	0.693	1074.262	0.172
	Unaware	0.757	0.683	522.355	0.096
	CD	0.723	0.432	-	-
	CR	0.747	0.554	192.893	0.092
	CFR	0.755	0.672	299.696	0.063
	CLAIRE	0.740	0.569	107.425	0.052
	MultiCFL	0.748	0.663	105.637	0.061
	CIRCUS	0.768	0.705	63.740	0.031

constraints with the fairness-utility tradeoff. We fix the local generation parameter to $l_{num} = 100$, ensuring sufficient local density for correcting decision boundaries without incurring excessive computational costs or redundancy. For the global generation parameter g_{num} , we perform a search over the set $\{50, 100, 300, 500, 800\}$. The optimal value is selected to maximize fairness improvement subject to maintaining competitive classification performance. For a detailed list of optimal g_{num} for each dataset and model, please refer to Table S(II) in the supplementary file.

The performance results of these methods on the DNN and ResNet classifiers are shown in Tables VI and VII, respectively. The detailed results of the Wilcoxon signed-rank test are provided in Table S(VII) of the supplementary file. We can see that UF exhibits higher MMD_L and MMD_K values due to the lack of mitigation measures for counterfactual discrimination. Interestingly, Unaware achieves the best MMD_L and MMD_K values on the Credit dataset, which can be attributed to its causal graph containing only a single descendant of the sensitive feature, thereby limiting the propagation of implicit bias. In contrast, Unaware's improvement in counterfactual

TABLE VII
EXPERIMENTAL RESULTS OF THE COUNTERFACTUAL FAIRNESS
ENHANCEMENT METHOD FOR THE RESNET

Dataset	Method	Metric			
		ACC	F1-score	MMD _L	MMD _K
Adult	UF	0.825	0.631	3086.336	0.499
	Unaware	0.819	0.621	1800.713	0.299
	CD	0.746	0.000	-	-
	CR	0.748	0.057	1076.554	0.278
	CFR	0.811	0.617	1059.337	0.282
	CLAIRE	0.769	0.264	383.538	0.175
	MultiCFL	0.777	0.508	379.375	0.174
	CIRCUS	0.795	0.611	166.877	0.090
Credit	UF	0.866	0.871	21.843	0.003
	Unaware	0.836	0.845	0.071	0.000
	CD	0.823	0.837	-	-
	CR	0.851	0.861	5.161	0.002
	CFR	0.835	0.862	5.156	0.001
	CLAIRE	0.858	0.865	0.820	0.001
	MultiCFL	0.853	0.867	0.377	0.000
	CIRCUS	0.873	0.872	0.260	0.000
Bail	UF	0.657	0.454	78.534	0.043
	Unaware	0.647	0.425	12.707	0.002
	CD	0.642	0.144	-	-
	CR	0.656	0.351	30.745	0.025
	CFR	0.641	0.446	4.165	0.001
	CLAIRE	0.645	0.439	10.721	0.009
	MultiCFL	0.645	0.451	54.458	0.031
	CIRCUS	0.654	0.458	2.113	0.001
Nutrition	UF	0.734	0.814	2730.514	0.268
	Unaware	0.724	0.801	596.245	0.120
	CD	0.697	0.748	-	-
	CR	0.717	0.804	16.420	0.060
	CFR	0.721	0.775	131.527	0.030
	CLAIRE	0.723	0.814	7.959	0.044
	MultiCFL	0.730	0.812	24.593	0.015
	CIRCUS	0.747	0.832	5.112	0.008
Average	UF	0.771	0.693	1479.307	0.203
	Unaware	0.757	0.673	602.434	0.105
	CD	0.727	0.432	-	-
	CR	0.743	0.518	282.220	0.091
	CFR	0.752	0.675	300.046	0.079
	CLAIRE	0.749	0.596	100.760	0.057
	MultiCFL	0.751	0.660	114.701	0.055
	CIRCUS	0.767	0.693	43.591	0.025

fairness is limited on the Adult and Nutrition datasets, where the causal models include multiple descendants of the sensitive feature, leading to implicit counterfactual discrimination. CD achieves complete counterfactual fairness by excluding sensitive feature descendants, but this results in a significant decline in classification performance across all datasets. CR strikes a balance between counterfactual fairness and classification performance on the Nutrition and Credit datasets, but performs poorly on the Adult dataset with a significant decline in classification performance. Additionally, CFR performs well on the Credit, Bail, and Nutrition datasets, showing only modest improvement in counterfactual fairness on the Adult dataset. On average, compared to CR and Unaware, CFR achieves a more substantial improvement in counterfactual fairness with relatively smaller sacrifices in classification performance. CLAIRE ranks second on average in the MMD_K metric with the DNN model and second on average in the MMD_L metric with the ResNet model, and performs well in improving fairness across all datasets. However, similar to CR, CLAIRE's fairness improvements come at the cost of substantially reduced classification performance. These results

underscore the challenge of enhancing counterfactual fairness through constraints, particularly the difficulty of achieving an appropriate balance between fairness and classification performance. As a counterfactual augmentation method, MultiCFL is similar to CFR, achieving competitive fairness while incurring a relatively small loss in classification performance. It ranks second in MMD_L with the DNN model and second in MMD_K with the ResNet model. This is attributed to MultiCFL's division of sensitive attributes into macro and individual levels, enabling the generated counterfactually fair samples to capture diverse causal dependencies and effectively mitigate the propagation of discrimination.

The proposed CIRCUS framework performs exceptionally well. On the DNN model, compared to UF, CIRCUS achieves average reductions of 94.1% in MMD_L and 82.0% in MMD_K, respectively, while outperforming the second-best results by an average of 39.7% and 40.4%. On the ResNet model, CIRCUS reduces MMD_L and MMD_K by an average of 97.1% and 87.7% compared to UF, and outperforms the second-best results by 56.7% and 54.5%. These results demonstrate the outstanding performance of CIRCUS in enhancing counterfactual fairness. Furthermore, on the DNN model, the ACC of CIRCUS is only 0.01 lower than that of the UF on average, with a slight improvement in the *F1*-score. On the ResNet model, CIRCUS's average ACC is 0.04 lower than that of UF, while its *F1*-score is tied with UF for the highest. These results further confirm that CIRCUS can significantly enhance the counterfactual fairness of a trained classifier while preserving classification performance. Additionally, CIRCUS ranks first in fairness metrics across all datasets except for the Credit dataset. On the Credit dataset, CIRCUS ranks second in the MMD_L but ties for first in MMD_K with Unaware, both achieving a score of 0.000. Moreover, CIRCUS achieves the highest ACC and *F1*-score on the Credit and Nutrition datasets, highlighting its strong generalization ability across both DNN classifiers and ResNet classifiers.

In summary, CIRCUS effectively identifies and corrects sources of counterfactual discrimination by performing causal interventions on key features during both the global and local stages, thereby enhancing the counterfactual fairness of the classifier. Combined with label bias correction for CDSs, CIRCUS significantly improves fairness while maintaining high classification performance, distinguishing itself from many constraint-based methods that compromise utility.

In the CIRCUS framework, setting different values of g_{num} finally generates counterfactually fair sample sets of different sizes. Using the Adult dataset as an example, Fig. 6 shows the trend in classifier performance and counterfactual fairness metrics after retraining the classifier with counterfactual fairness samples of different sizes. As shown in the figure, increasing g_{num} (i.e., more counterfactually fair samples) leads to a gradual decrease in the ACC of the DNN and ResNet classifiers, while significantly improving counterfactual fairness. The substantial decrease in MMD_L and MMD_K provides direct evidence of this improvement in counterfactual fairness. These results indicate that the CIRCUS framework effectively enhances counterfactual fairness through bias correction pre-processing while also validating our claim that this approach

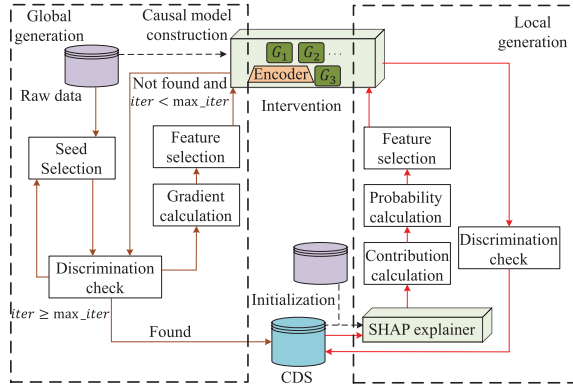


Fig. 5. Overview of the process for generating CDSs in the CIRCUS framework.

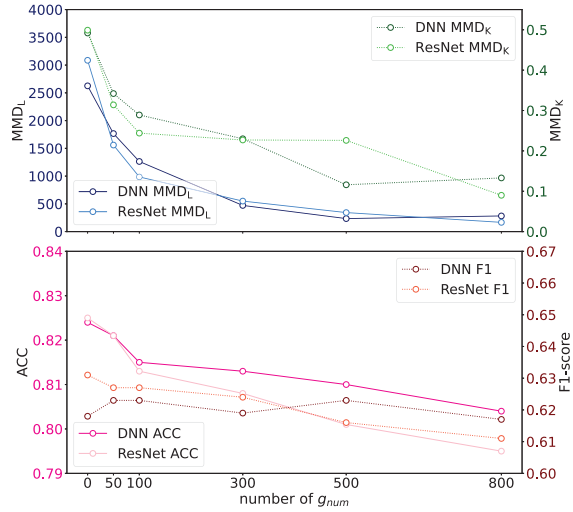


Fig. 6. Trend of metric changes in the CIRCUS framework on the Adult dataset.

does not significantly degrade classifier performance. Notably, as indicated in Table S(II) of the supplementary file, complex models like ResNet typically require higher g_{num} settings to sufficiently cover the decision space, thereby necessitating a larger volume of counterfactually fair samples to fully maximize fairness improvements.

VII. CONCLUSION AND FUTURE WORK

In this article, we first propose the CITGAN, a model that effectively captures and fits the underlying causal structure in tabular data and generates counterfactual samples corresponding to different values of the sensitive feature for each instance. Building on CITGAN, we design an innovative CIRCUS framework, which generates CDSs guided by gradients and feature contributions, and transforms them into counterfactually fair samples through label bias correction preprocessing. Experimental results show that CITGAN outperforms existing causal model fitting methods in data generation quality, and that the CIRCUS framework enhances counterfactual fairness while maintaining strong classification performance. Together, CITGAN and CIRCUS provide a systematic tool for counterfactual reasoning and fairness enhancement in trustworthy AI.

Currently, the CIRCUS framework intervenes on features by combining loss-function gradients with feature contributions.

It is primarily designed for differentiable classifiers, where smooth decision boundaries ensure the effectiveness of the interventions. In future work, this framework may be extended to nondifferentiable models or models with nonsmooth decision boundaries (e.g., decision trees). Additionally, our method can be further optimized by integrating advanced tensor-based learning techniques [38], [39] to enhance feature discrimination and improve the model's expressiveness. Furthermore, we plan to employ intelligent optimization algorithms, such as P2SO [40], PGA-PSO [41], and SPSO [42], for adaptive hyperparameter tuning, aiming to enhance the model's robustness and efficiency.

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